

Ejercicio VirtualBox I

1. Direccionamiento IP

- Web

- Puente: 192.168.0.117
- Red interna: 10.10.10.10

```
user@serverweb:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f9:db:18 brd ff:ff:ff:ff:ff:ff
    altname enx080027f9db18
    inet 192.168.0.117/24 brd 192.168.0.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83085sec preferred_lft 72285sec
    inet6 fe80::eabf:e30b:3fe3:d95e/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:6a:3a:80 brd ff:ff:ff:ff:ff:ff
    altname enx0800276a3a80
    inet 10.10.10.10/24 brd 10.10.10.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe6a:3a80/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
user@serverweb:~$ _
```

- Base de datos

- Red interna: 10.10.10.20

```
user@serverDB:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:02:51:0b brd ff:ff:ff:ff:ff:ff
    altname enx08002702510b
    inet 10.10.10.20/24 brd 10.10.10.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe02:510b/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
user@serverDB:~$ <
```

- Sistema de archivos
 - Red interna: 10.10.10.30

```
user@serverFS:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0c:89:f4 brd ff:ff:ff:ff:ff:ff
    altname enx0800270c89f4
    inet 10.10.10.30/24 brd 10.10.10.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe0c:89f4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
user@serverFS:~$
```

2. Tarjetas de red

- Web
 - 2 tarjetas de red
 - Puente y Red interna

General

Nombre: ServerWeb
Sistema operativo: Debian (64-bit)

Sistema

Memoria base: 1024 MB
Orden de arranque: Disquete, Óptica, Disco duro
Aceleración: Paginación anidada, Paravirtualización KVM

Pantalla

Memoria de vídeo: 16 MB
Controlador gráfico: VMSVGA
Servidor de escritorio remoto: Inhabilitado
Grabación: Inhabilitado

Almacenamiento

Controlador: IDE
Dispositivo IDE primario 0: [Unidad óptica] Vacío
Controlador: SATA
Puerto SATA 0: ServerWeb.vdi (Normal, 5.00 GB)

Audio

Controlador de anfitrión: Predeterminado
Controlador: ICH AC97

Red

Adaptador 1: Intel PRO/1000 MT Desktop (Adaptador puente, «wlp58s0»)
Adaptador 2: Intel PRO/1000 MT Desktop (Red interna, «backend-net»)

- Base de datos

- 1 tarjeta de red: Red interna

General	
Nombre:	ServerDB
Sistema operativo:	Debian (64-bit)
Sistema	
Memoria base:	1024 MB
Procesadores:	2
Orden de arranque:	Disquete, Óptica, Disco duro
Aceleración:	Paginación anidada, Paravirtualización KVM
Pantalla	
Memoria de vídeo:	16 MB
Controlador gráfico:	VMSVGA
Servidor de escritorio remoto:	Inhabilitado
Grabación:	Inhabilitado
Almacenamiento	
Controlador:	IDE
Dispositivo IDE primario 0:	[Unidad óptica] Vacío
Controlador:	SATA
Puerto SATA 0:	ServerDB.vdi (Normal, 5.00 GB)
Audio	
Controlador de anfitrión:	Predeterminado
Controlador:	ICH AC97
Red	
Adaptador 2:	Intel PRO/1000 MT Desktop (Red interna, «backend-net»)

- Sistema de archivos

- 1 tarjeta de red: Red interna

General	
Nombre:	ServerFS
Sistema operativo:	Debian (64-bit)
Sistema	
Memoria base:	1024 MB
Orden de arranque:	Disquete, Óptica, Disco duro
Aceleración:	Paginación anidada, Paravirtualización KVM
Pantalla	
Memoria de vídeo:	16 MB
Controlador gráfico:	VMSVGA
Servidor de escritorio remoto:	Inhabilitado
Grabación:	Inhabilitado
Almacenamiento	
Controlador:	IDE
Dispositivo IDE primario 0:	[Unidad óptica] Vacío
Controlador:	SATA
Puerto SATA 0:	ServerFS.vdi (Normal, 5.00 GB)
Audio	
Controlador de anfitrión:	Predeterminado
Controlador:	ICH AC97
Red	
Adaptador 2:	Intel PRO/1000 MT Desktop (Red interna, «backend-net»)

3.Pruebas de Comunicación

- Web

- Ping a BD y FS

```
user@serverweb:~$ ping 10.10.10.20
PING 10.10.10.20 (10.10.10.20) 56(84) bytes of data.
64 bytes from 10.10.10.20: icmp_seq=1 ttl=64 time=4.82 ms
64 bytes from 10.10.10.20: icmp_seq=2 ttl=64 time=1.58 ms
64 bytes from 10.10.10.20: icmp_seq=3 ttl=64 time=1.12 ms
64 bytes from 10.10.10.20: icmp_seq=4 ttl=64 time=1.21 ms
^C
--- 10.10.10.20 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.120/2.181/4.818/1.531 ms
user@serverweb:~$ ping 10.10.10.30
PING 10.10.10.30 (10.10.10.30) 56(84) bytes of data.
64 bytes from 10.10.10.30: icmp_seq=1 ttl=64 time=3.36 ms
64 bytes from 10.10.10.30: icmp_seq=2 ttl=64 time=1.01 ms
64 bytes from 10.10.10.30: icmp_seq=3 ttl=64 time=1.14 ms
64 bytes from 10.10.10.30: icmp_seq=4 ttl=64 time=1.20 ms
^C
--- 10.10.10.30 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.007/1.674/3.355/0.972 ms
```

- Conexión a BD

```
user@serverweb:~$ mysql -h 10.10.10.20 -u webuser -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g
Your MariaDB connection id is 41
Server version: 11.8.6-MariaDB-0+deb13u1 from Debian

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show database;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'atabase' at line 1
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| practica |
+-----+
2 rows in set (0.007 sec)
```

```

MariaDB [(none)]> use practica;
Database changed
MariaDB [practica]> create table dinosaurios (id INT PRIMARY KE
Query OK, 0 rows affected (0.039 sec)

MariaDB [practica]> show tables;
+-----+
| Tables_in_practica |
+-----+
| dinosaurios        |
+-----+
1 row in set (0.004 sec)

MariaDB [practica]> insert into dinosaurios(nombre,tipo)
-> VALUES
-> ('T-Rex', 'Carnivoro')
-> ('Triceratops', 'Herbivoro')
-> ('Brachiosaurio', 'Herbivoro');
ERROR 1064 (42000): You have an error in your SQL syntax; check
riceratops', 'Herbivoro')
('Brachiosaurio', 'Herbivoro')' at line 4
MariaDB [practica]> insert into dinosaurios(nombre,tipo) VALUES
Query OK, 3 rows affected (0.009 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [practica]> show tables;
+-----+
| Tables_in_practica |
+-----+
| dinosaurios        |
+-----+
1 row in set (0.007 sec)

MariaDB [practica]> select * from dinosarios
-> ;
ERROR 1146 (42S02): Table 'practica.dinosarios' doesn't exist
MariaDB [practica]> select * from dinosaurios
-> ;
+-----+-----+-----+
| id | nombre        | tipo        |
+-----+-----+-----+
| 1  | T-Rex         | Carnivoro   |
| 2  | Triceratops   | Herbivoro   |
| 3  | Brachiosaurio | Herbivoro   |
+-----+-----+-----+
3 rows in set (0.003 sec)

```

- Conexion a sistema de archivos

```

user@serverweb:~$ curl http://10.10.10.30/imagenes/imagen.jpg --output test.jpg
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total   Spent    Left   Speed
100 86063  100 86063    0     0  3322k      0  --:--:-- --:--:-- --:--:-- 3361k
user@serverweb:~$ ls
index.php  prueba.jpg  test.jpg

```

- Base de datos

- Ping a Web

```
user@serverDB:~$ ping 10.10.10.10
PING 10.10.10.10 (10.10.10.10) 56(84) bytes of data.
64 bytes from 10.10.10.10: icmp_seq=1 ttl=64 time=2.43 ms
64 bytes from 10.10.10.10: icmp_seq=2 ttl=64 time=1.23 ms
64 bytes from 10.10.10.10: icmp_seq=3 ttl=64 time=1.07 ms
64 bytes from 10.10.10.10: icmp_seq=4 ttl=64 time=1.02 ms
^C
--- 10.10.10.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.015/1.435/2.432/0.580 ms
```

- Sistema de archivos

- Ping a Web

```
user@serverFS:~$ ping 10.10.10.10
PING 10.10.10.10 (10.10.10.10) 56(84) bytes of data.
64 bytes from 10.10.10.10: icmp_seq=1 ttl=64 time=3.49 ms
64 bytes from 10.10.10.10: icmp_seq=2 ttl=64 time=0.995 ms
64 bytes from 10.10.10.10: icmp_seq=3 ttl=64 time=1.07 ms
64 bytes from 10.10.10.10: icmp_seq=4 ttl=64 time=1.13 ms
^C
--- 10.10.10.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.995/1.672/3.491/1.051 ms
```

- Maquina local

```
> ping 192.168.0.117
PING 192.168.0.117 (192.168.0.117) 56(84) bytes of data.
64 bytes from 192.168.0.117: icmp_seq=1 ttl=64 time=1.47 ms
64 bytes from 192.168.0.117: icmp_seq=2 ttl=64 time=0.583 ms
64 bytes from 192.168.0.117: icmp_seq=3 ttl=64 time=0.659 ms
64 bytes from 192.168.0.117: icmp_seq=4 ttl=64 time=0.627 ms
^C
--- 192.168.0.117 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3058ms
rtt min/avg/max/mdev = 0.583/0.835/1.472/0.368 ms
```

- Prueba desde el navegador



Servicio Web 2 Capas

ID	Nombre	Tipo
1	T-Rex	Carnivoro
2	Triceratops	Herbivoro
3	Brachiosaurio	Herbivoro

Imagen desde servidor de archivos

